Microbiology

RABIES

OBJECTIVES

At the end of this lecture the students will be able to:

1- Define rabies, its epidemiology, mode of transmission.

2- Describe symptoms ,diagnosis , treatment , prognosis

3- Discus its preventive measures both in human and animal .

RABIES VIRUS

Rabies is an acute infection of the central nervous system that is almost always fatal.

The virus is usually transmitted to humans from the bite of a rabid animal.

Although the number of human cases is small, rabies is a major public health problem because it is widespread among animal reservoirs

Rabies virus



INTRODUCTION



Rabies is one of the oldest recognized diseases affecting humans and one of the most important zoonotic diseases Discovered and named by Girolamo Fracastoro in 15th century



Acute, highly infectious and fatal disease of the CNS that causes acute encephalitis (inflammation of the brain) in warm blooded animals Caused by Lyssavirus type 1 Transmitted by bites of rabid animal



The disease infects domestic and wild animals, and is spread to people through close contact with infected saliva via bites or scratches.



Dogs are the source of 99% of human rabies deaths Once symptoms of the disease develop, rabies is nearly always fatal.

INTRODUCTION

- Painting of a rabid dog biting a man Arabic A.D. 1224 Baghdad school, by Abdallah ibn al- Fadl
- Picture courtesy of Smithonian Institution, Washington D.C



PROPERTIES OF THE RABIES VIRUS

The viruses are classified in the family Rhabdoviridae involves wide array of viruses with broad host range.

Group includes the deadly rabies virus The rhabdoviruses are rod- or bullet-shaped particles measuring 75 x 180 nm.

The particles are surrounded by a membranous envelope with protruding spikes, which are composed of the viral glycoprotein.

Inside the envelope is a ribonucleocapsid.

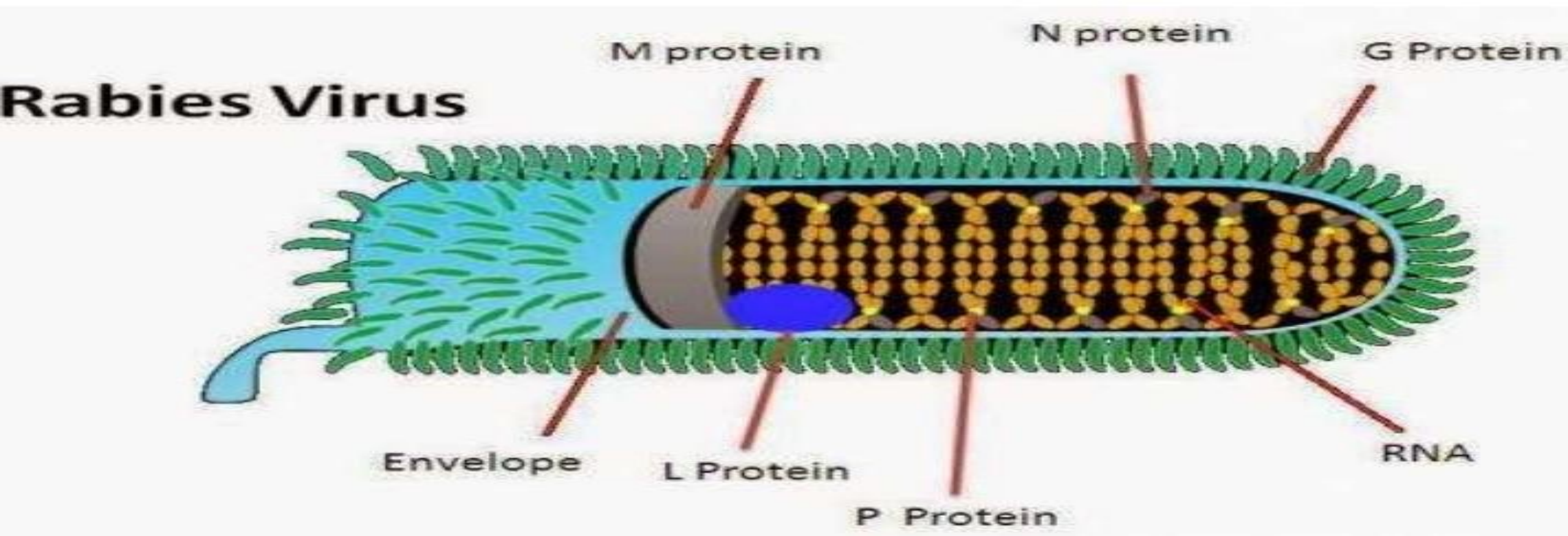
PROPERTIES OF THE RABIES VIRUS

The genome is single-stranded, negative-sense RNA Virions contain an RNA-dependent RNA polymerase.

Replication: Cytoplasm; virions bud from plasma membrane There is a single serotype of rabies virus.

The G glycoprotein is a major factor in rabies virus neuroinvasiveness and pathogenicity

PROPERTIES OF THE RABIES VIRUS



RABIES VIRUS REPLICATION

Rabies virus attaches to cells via its glycoprotein spikes; acetylcholine receptor may serve as a cellular receptor for rabies virus.

The single-stranded RNA genome is transcribed by the virion RNA polymerase to five mRNA species. The template for transcription is the genome RNA in the form of ribonucleoprotein (RNP) (encased in N protein and containing the viral transcriptase).

The mRNAs code for the five virion proteins: nucleocapsid (N), polymerase proteins (L, P), matrix (M), and glycoprotein (G).

The genome RNP is a template for complementary positive-sense RNA, which is responsible for the generation of negative-sense progeny RNA.

Ongoing translation is required for replication, particularly of viral N and P proteins.

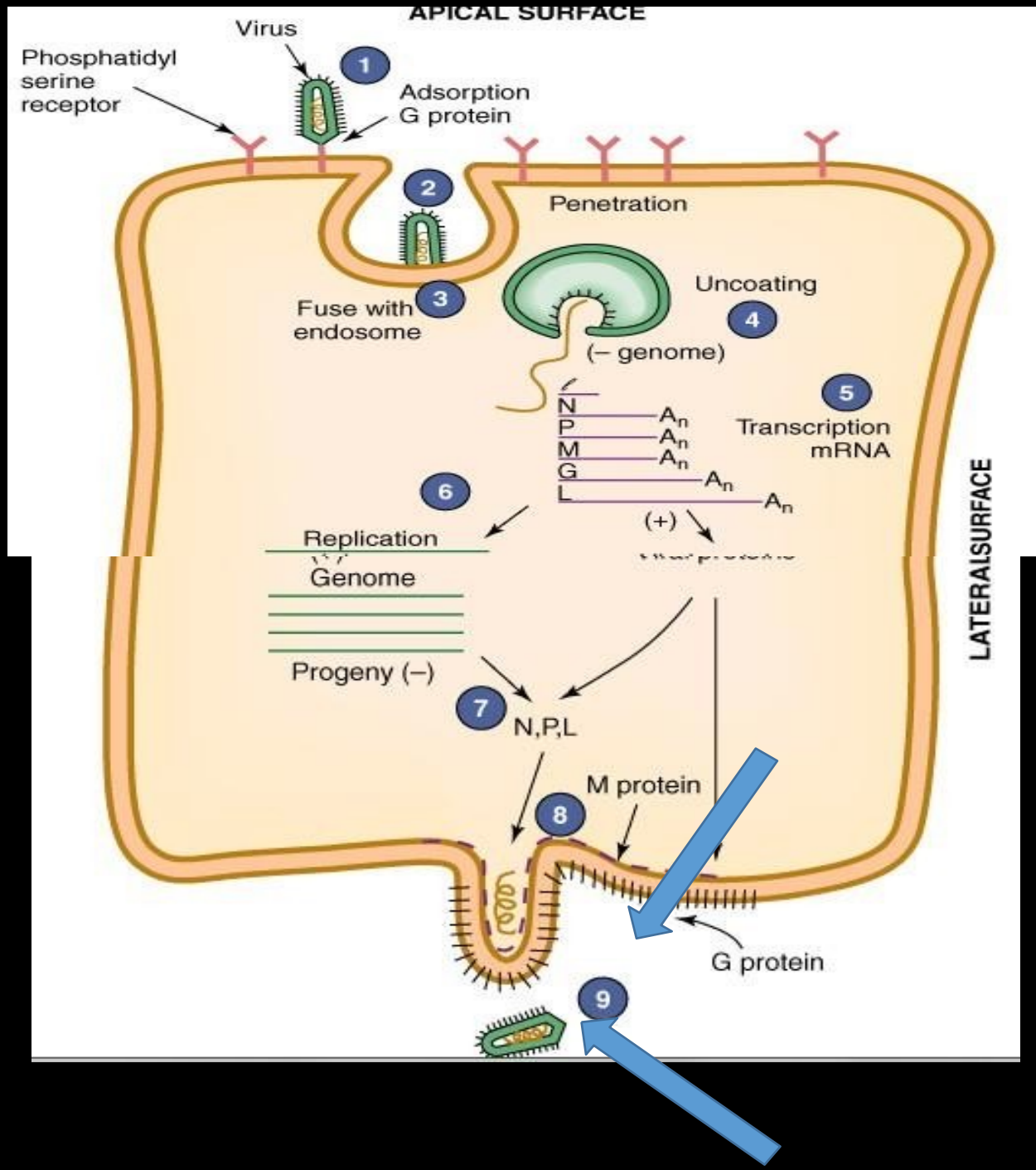
RABIES VIRUS REPLICATION

The newly replicated genomic RNA associates with the viral transcriptase and nucleoprotein to form RNP cores in the cytoplasm.

The particles acquired an envelope by budding through the plasma membrane.

The viral M matrix protein forms a layer on the inner side of the envelope, whereas the viral G glycoprotein is on the outer layer and forms the spikes

RABIES VIRUS REPLICATION



HOST & REsERVOZR





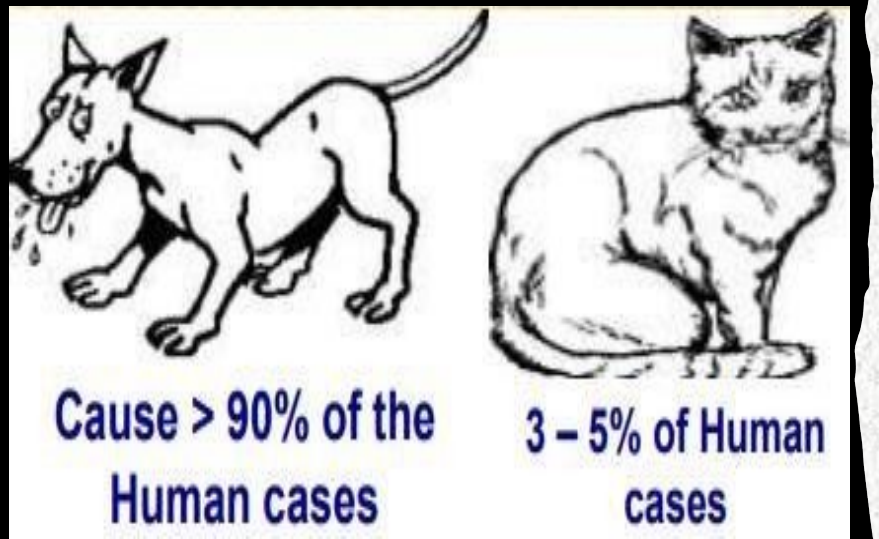








SOURCE OF INFECTION



- The source of infection to man is the saliva of rabid animals.
- In dogs & cats, the virus may be present in the saliva for 3-4 days before the clinical onset & during the course of illness till death.
- Inhalation of virus containing aerosols.
- Less infectious materials are other body fluids like serum , urine and milk
- In humans-saliva ,sweat ,semen and tears contain the virus.
- Cornea of human cases constitutes an infective material.(rabies antigen detected in corneal cells)

INCUBATION PERIOD



It varies from 3 weeks to 3 months(15 days to 1 year). weeks following exposure.



The closer the bite to the brain, the shorter the incubation.



Rabies virus travels 1 cm per day

FACTORS INFLUENCING THE INCUBATION PERIOD

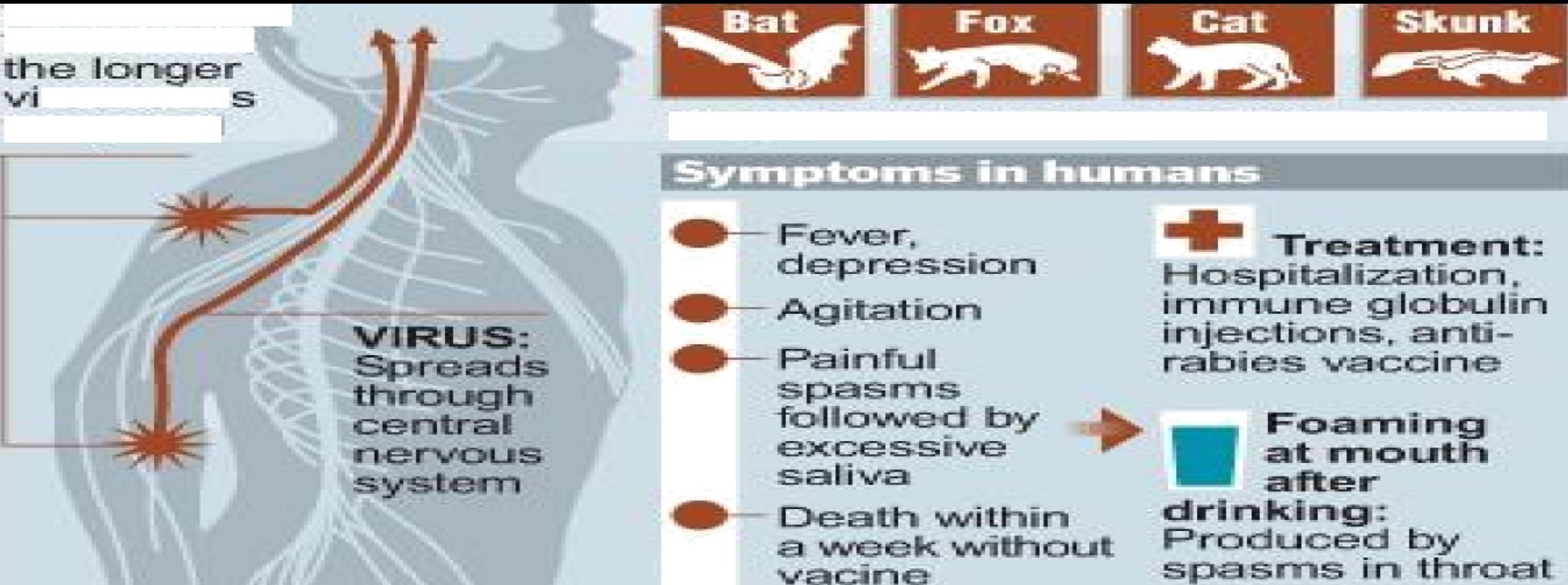
1. Site Of The Bite.
2. Severity Of The Bite.
3. Species Of The Biting Animal.
4. Richness Of The Nerve Supply.
5. Amount Of Saliva Deposited.
6. Protection Through Clothes.
7. Partial Treatment Taken .

PERIOD OF INFECTIVITY

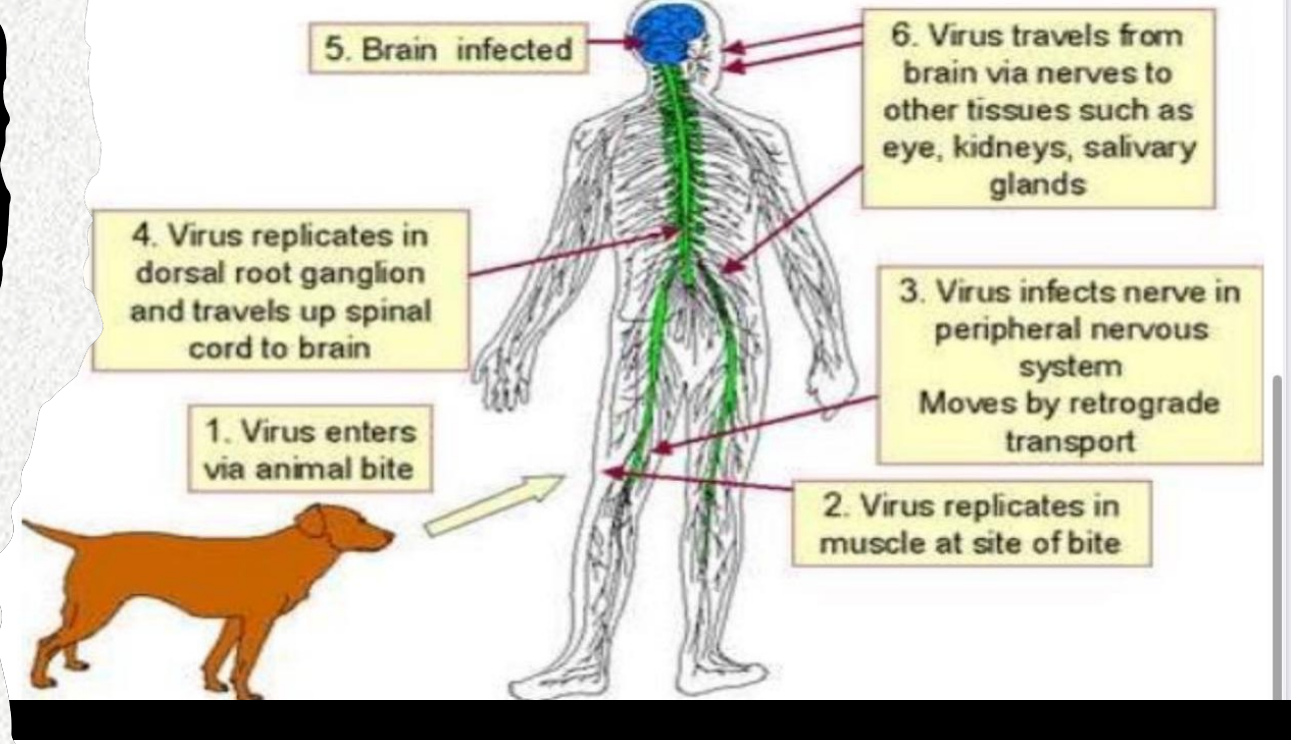
The rabid animal is infectious during the last 3 to 5 days of incubation period and also during the entire period of illness, which is about 8 to 10 days.

Thus the total period of infectivity is about 12 to 15 days.

RABIES VIRUS



Pathogenesis



Live virus → Epidermis, Mucus membrane

↓

Peripheral nerve

↓ centripetally

CNS (gray matter)

↓ centripetally

Other tissue (salivary glands,…)

Pathogenesis

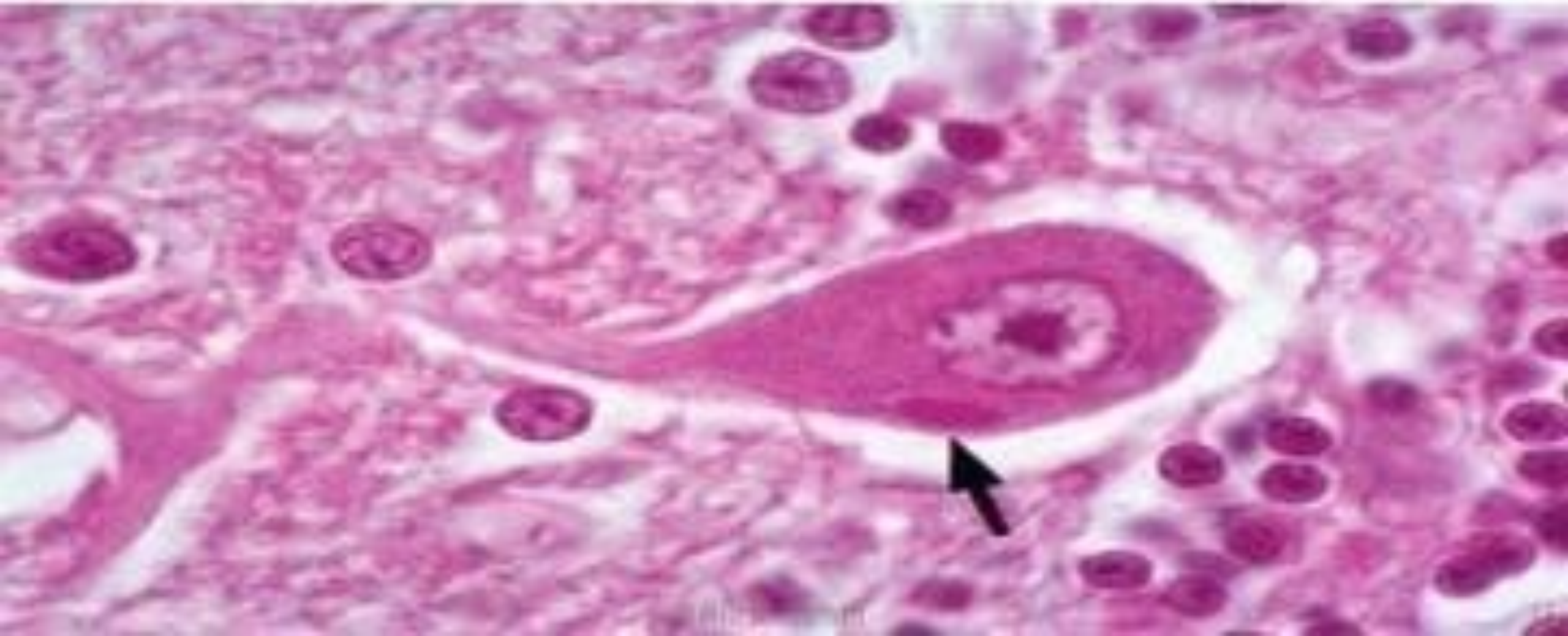
Rabies virus produces a specific eosinophilic cytoplasmic inclusion, the Negri body, in infected nerve cells.

Negri bodies are filled with viral nucleocapsids.

The presence of such inclusions is pathognomonic of rabies but is not observed in at least 20% of cases.

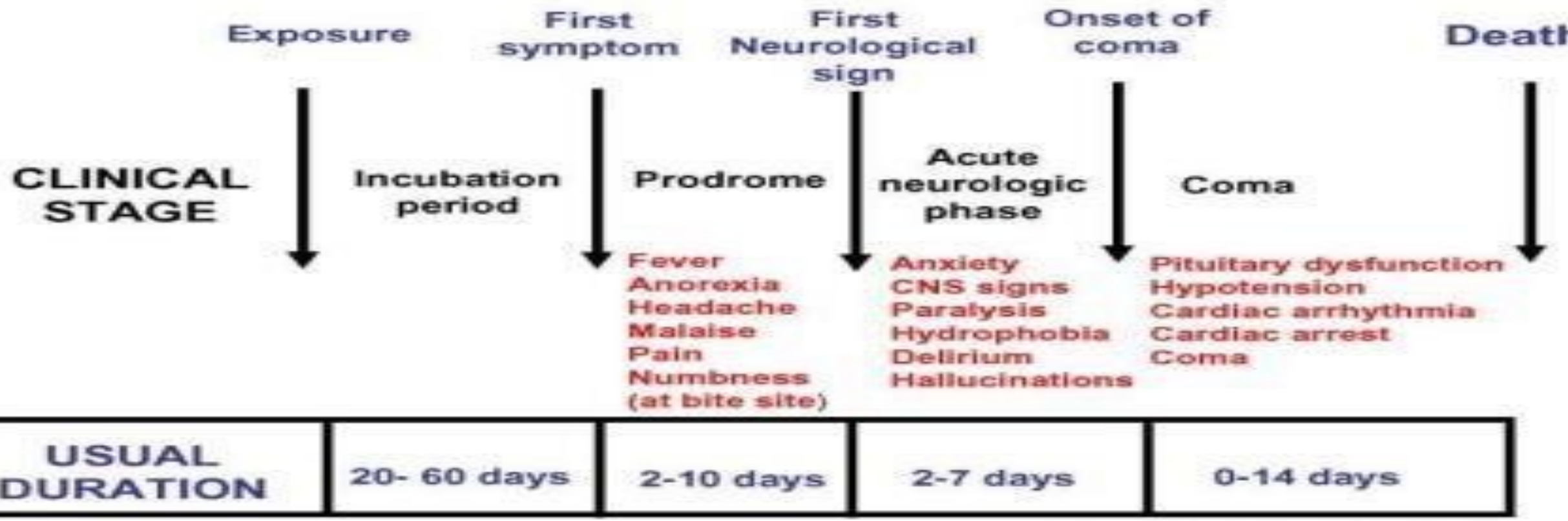
Therefore, the absence of Negri bodies does not rule out rabies as a diagnosis

NEGRI BODIES



RABIES VIRUS

Clinical Course of Rabies in Humans



Based on: Churchill Livingstone

CLINICAL FINDINGS

spread to humans by bites of rabid animals or by contact with saliva from rabid animals.

The disease is an acute, fatal encephalitis. The incubation period in humans is typically 1–2 months. It is usually shorter in children than in adults.

The clinical spectrum can be divided into three phases: a short prodromal phase, an acute neurologic phase, and coma. The prodromal phase may show any of the following nonspecific symptoms: malaise, anorexia, headache, photophobia, nausea and vomiting, sore throat, and fever

During the acute neurologic phase, which lasts 2–7 days, patients show signs of nervous system dysfunction such as nervousness, and hallucinations. A large fraction of patients will exhibit hydrophobia (fear of water) followed by convulsive seizures or coma and death. The major cause of death is respiratory paralysis.

LABORATORY DIAGNOSIS

1. Tissues infected with rabies virus are currently identified most rapidly and accurately by means of immunofluorescence or immunoperoxidase staining using antirabies monoclonal antibodies (IHC)

2. A definitive pathologic diagnosis of rabies can be based on the finding of Negri bodies in the brain or the spinal cord they have basophilic granules in an eosinophilic matrix.

3. Reverse transcription-PCR testing can be used to amplify parts of a rabies virus genome from fixed or unfixed brain tissue.

4. Viral Isolation: Available tissue is inoculated intracerebrally into suckling mice

5. Serology: Serum antibodies to rabies can be detected by immunofluorescence

PREVENTION TREATMENT & CONTROL

All vaccines for human use contain only inactivated rabies virus (Killed vaccine), with Human Rabies immune globulin (HRIG)

There is no successful treatment for clinical rabies.

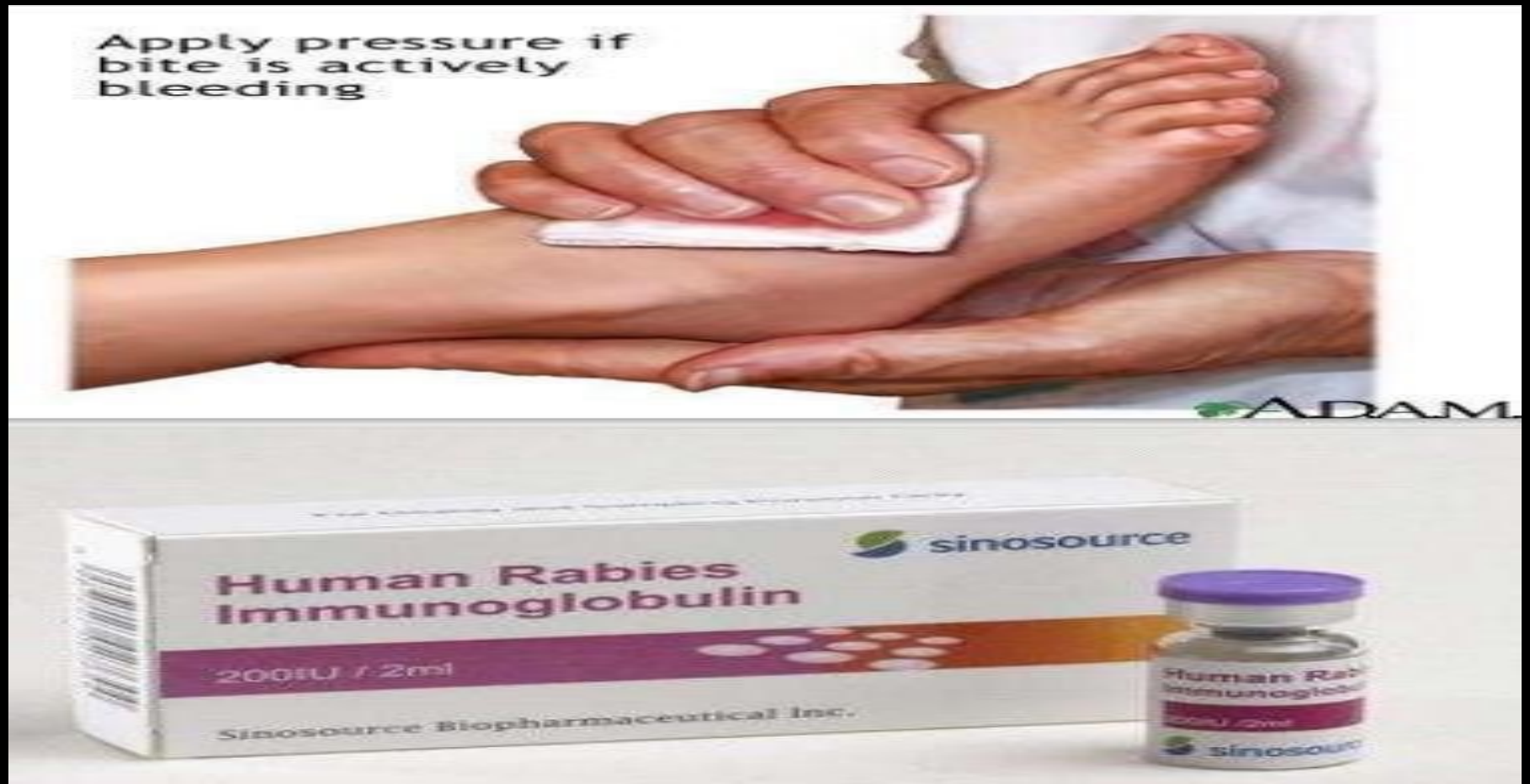
Preexposure vaccination is desirable for all persons who are at high risk of contact with rabid animals, such as veterinarians, animal care personnel, certain laboratory workers

An oral vaccinia–rabies glycoprotein recombinant virus vaccine (V-RG) proved effective at controlling rabies in foxes in Europe

PEP (POST EXPOSURE PROPHYLAXIS)

- Provided to subjects after rabies exposure.
- Consists of wound care, rabies immune globulin, and vaccine.
- Cleansing
- Chemical Treatment
- Anti-Rabies Serum
- Antibiotics & anti-tetanus measure
- Observe the animal for 10 days.
- Wash lesions well with soap and water (tetanus booster)
- Infiltrate rabies immune globulin (20 IU/kg) into and around the margin of the bites.
- Administer vaccine on days 0,3,7,14, and 28. (90)

PEP (POST EXPOSURE PROPHYLAXIS)



TREATMENT

- POST-EXPOSURE TREATMENT (PET)
- LOCAL WOUND TREATMENT
- Wash with soap/detergent and water preferably for 10 mins.
- Apply alcohol, povidone iodine/ any antiseptic
- Anti-Tetanus
- Avoid suturing wounds
- Don’t apply ointment, cream/ wound dressing
- ANTIMICROBIAL - Amoxicillin - Cloxacillin - Cefuroxime
- For those instances where there’s no obvious signs of infection(Amoxicillin as prophylaxis)
- Educate the public simple local wound treatment & warn not to use procedures that may further contaminate the wounds

TREATMENT







Initial treatment for an animal bite should include thorough cleansing however all animal bites should be seen by a physician

Prevention of human rabies

The 5 Important Things



Magico-Religious Practices *(e.g. witchcraft, turmeric powder etc.) DO NOT HELP*



Wash the wound thoroughly with plenty of water and soap



Apply an antiseptic (povidone iodine) or even alcohol



Do not cover or Suture the wound



Vaccinate Immediately e.g. *Rabipur 1 mL IM*

VACCINE ADMINISTRATION

1.Intramuscular Schedules

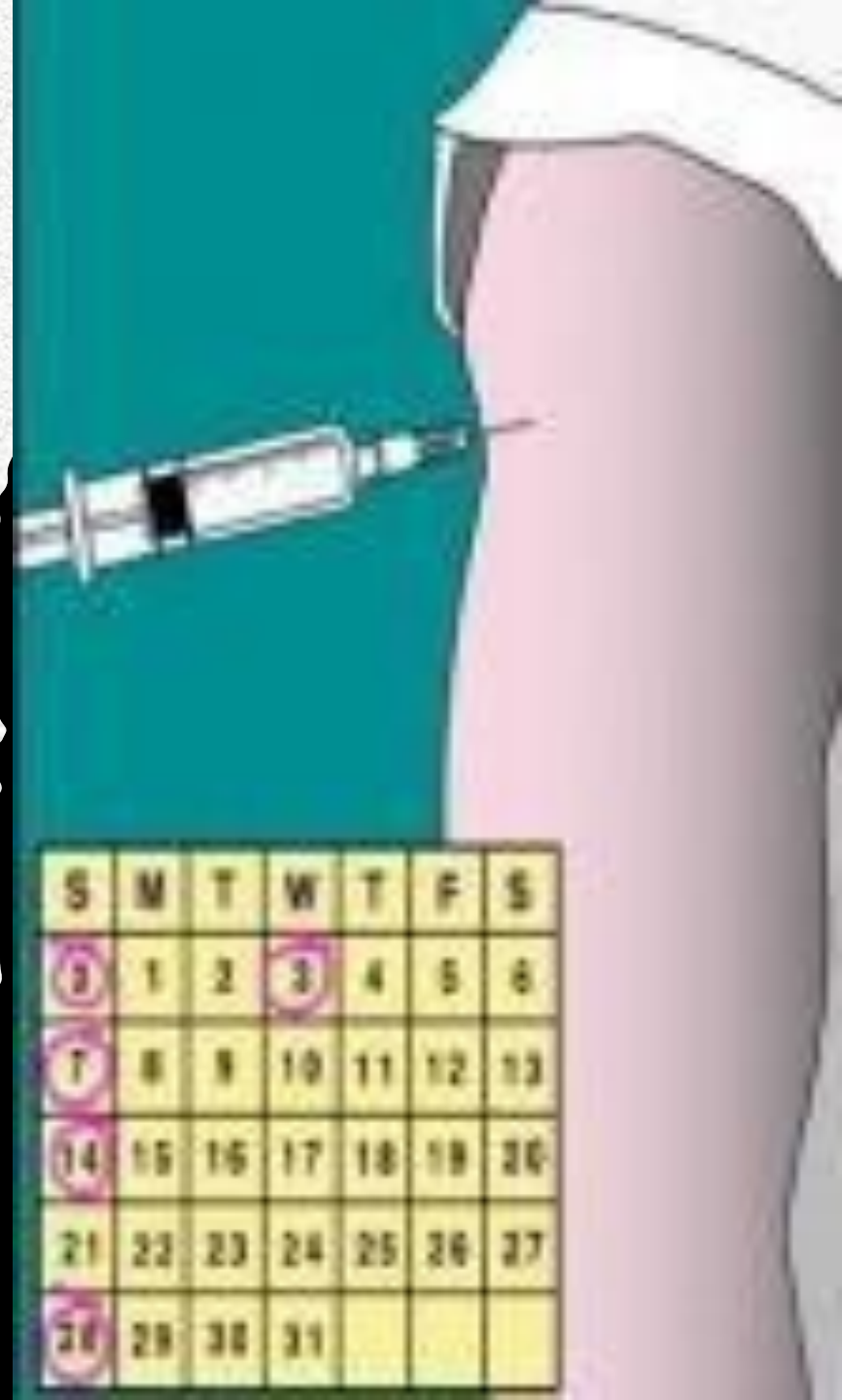
6 doses schedule

Reduced multisite intramuscular regimen (2-1-1)

2.intradermal Schedules

A.2-Site Intradermal schedule(2-2- 2-0-1-1)

B.8-Site intradermal schedule(8-0- 4-0-1-1)




Thank you